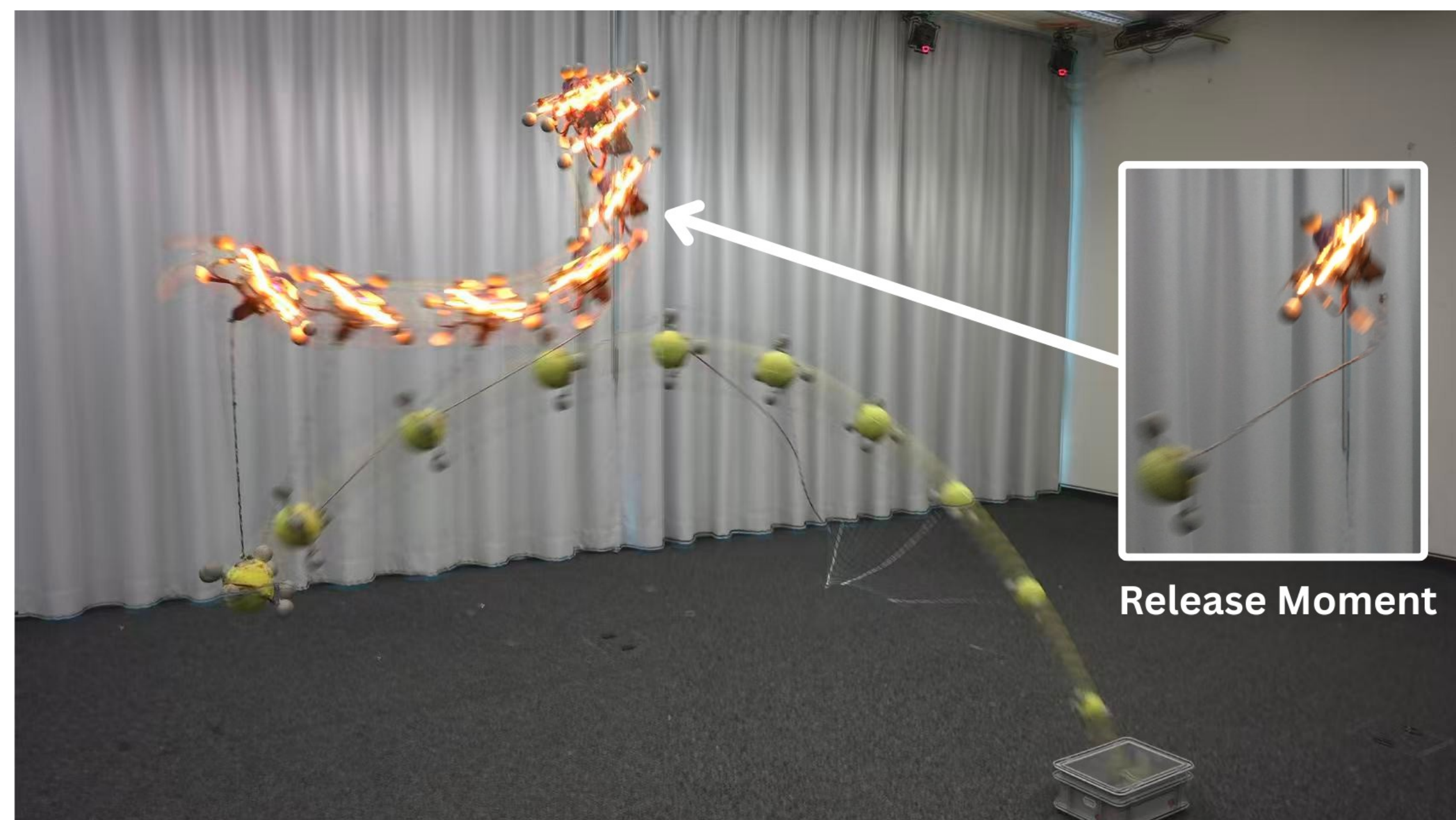


Motivation

Throwing a suspended payload delivers it faster than carrying it, but after release the payload flies ballistically: **small release errors become large landing errors**.

Model-based planners optimize on **simplified models** and hand the plan to a separate tracking controller. The release state deviates from the plan, and bounding that error forces **slow, conservative casts**.



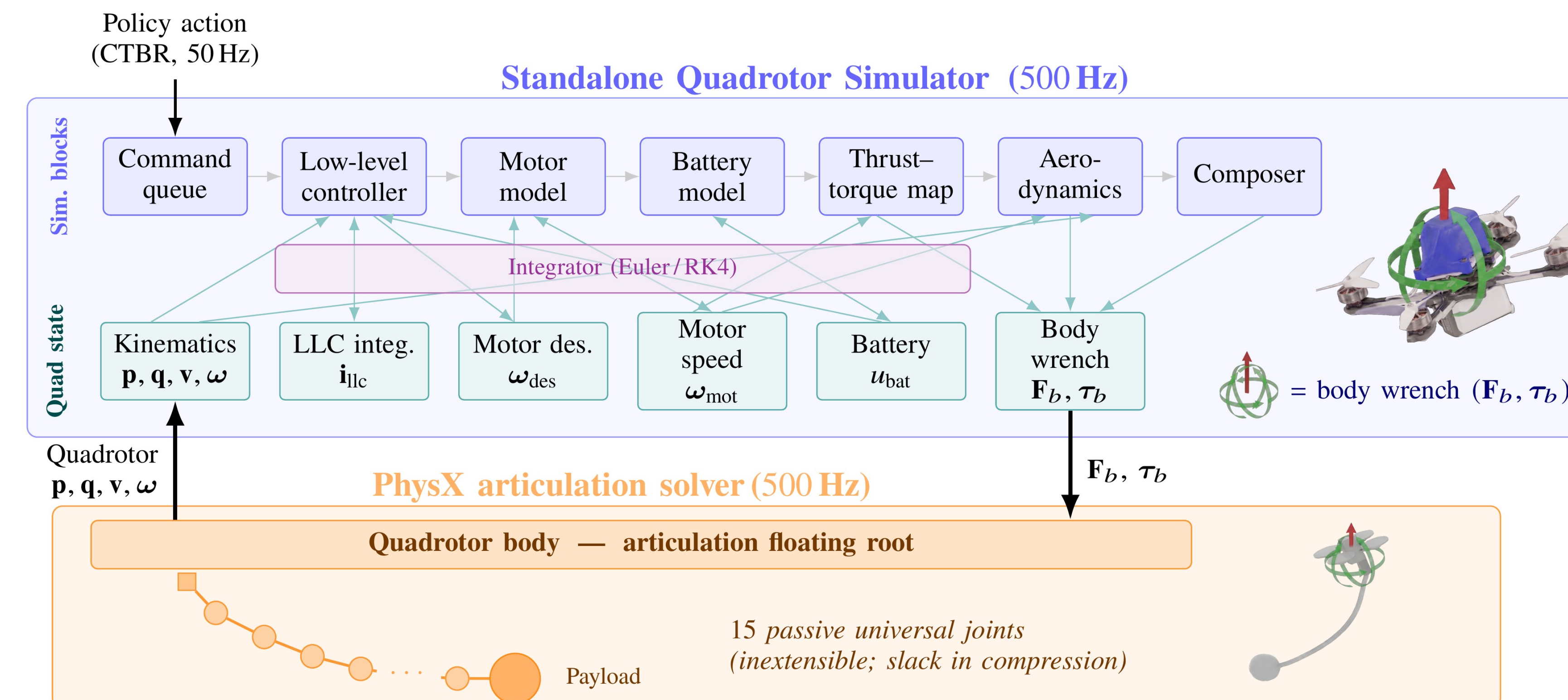
Contribution

- **Hybrid simulator**: an identified quadrotor model supplies the body wrench to a multibody solver that advances quadrotor, rope, and payload together.
- **Direct control**: an RL policy commands collective thrust, body rates, and the release, with no intermediate trajectory.
- **Zero-shot hardware throws** against two model-based methods, plus hardware ablations of each model component.

Key Takeaways

- Keeping the **identified quadrotor response** inside the coupled cable–payload simulation enables zero-shot throwing.
- Closed-loop CTBR control **uses more of the platform**: the fastest throws, at accuracy close to an error-bounded planner.
- The sim-to-real gap stays at ≈ 3.6 cm up to an out-of-distribution 2.5 m target.
- A policy on **visual features** (hardware-in-the-loop) lands within 10.1 cm on average.

Hybrid Simulation and Training



Hybrid simulator

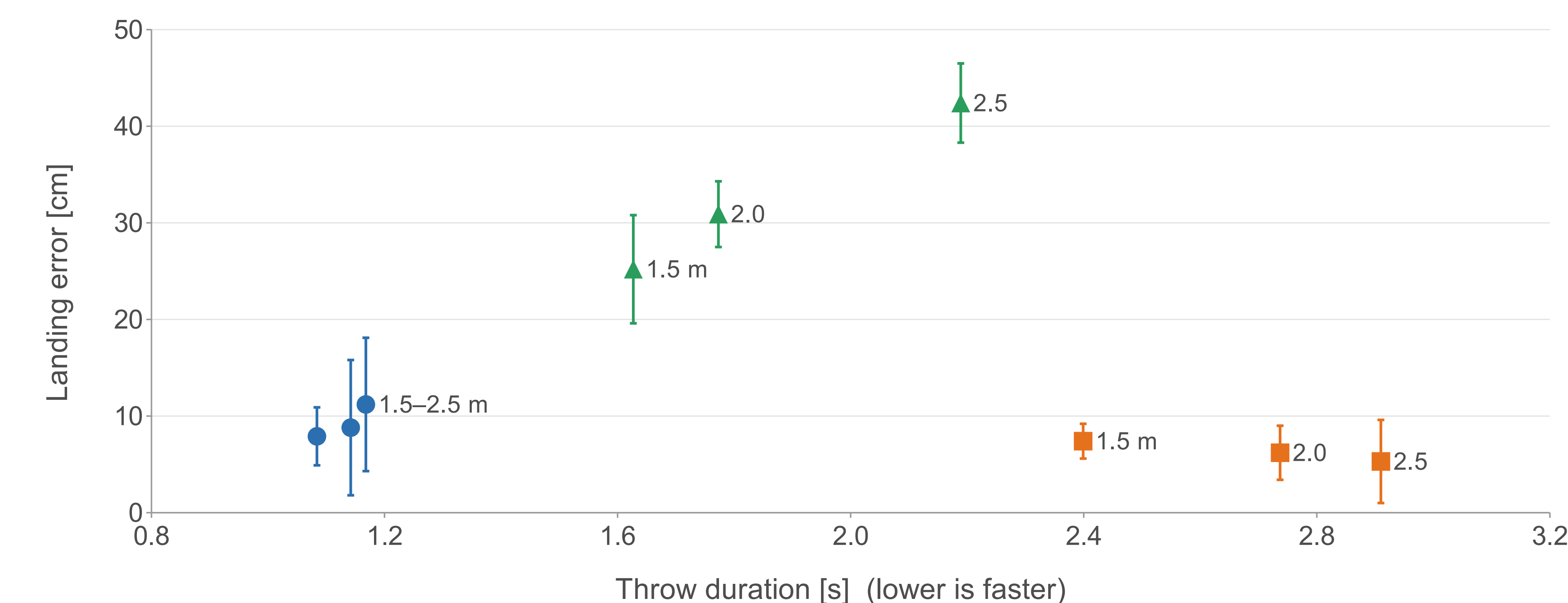
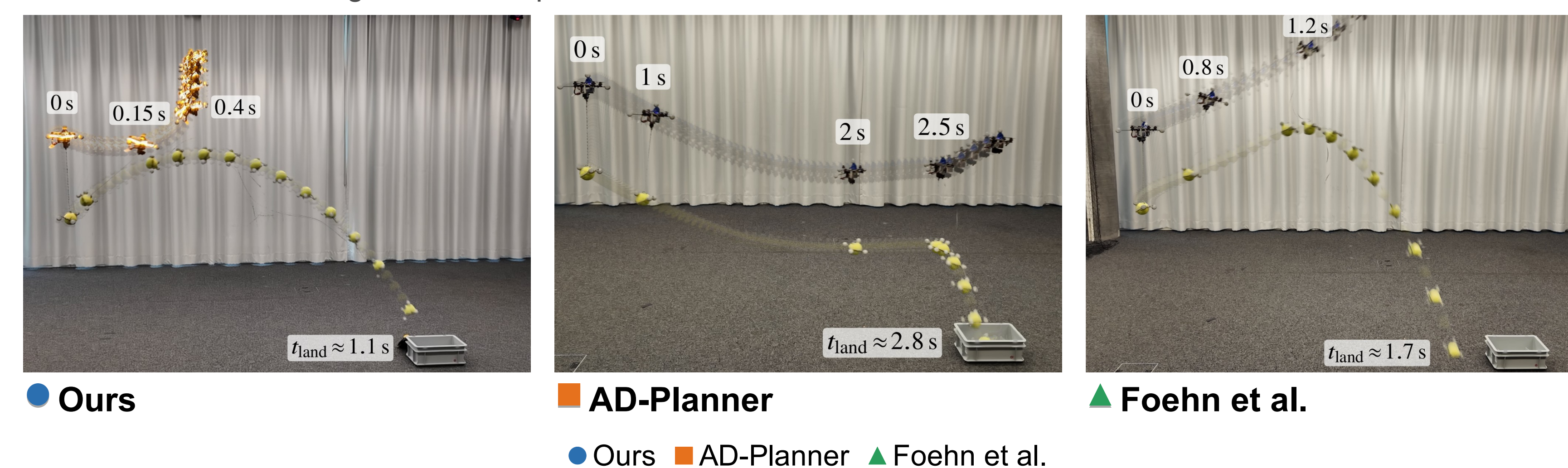
- Identified once on the standalone quadrotor: delay, rate controller, motors, battery, thrust map, aerodynamics.
- Only the body wrench and root kinematics are exchanged, at 500 Hz; a 15-segment chain lets the rope go slack.

RL training

- PPO with an asymmetric actor–critic; collective thrust, body rates, and release latch at 50 Hz.
- Reward on the predicted landing point and flight time; domain randomization; 70 min on one GPU.

Zero-Shot Hardware Throws

Throws to the 2.0 m target; timestamps from the start of the throw.



Mean $\pm$ std over 30 throws per target (ours) and 10 (baselines); labels give the target distance.

What Makes It Work

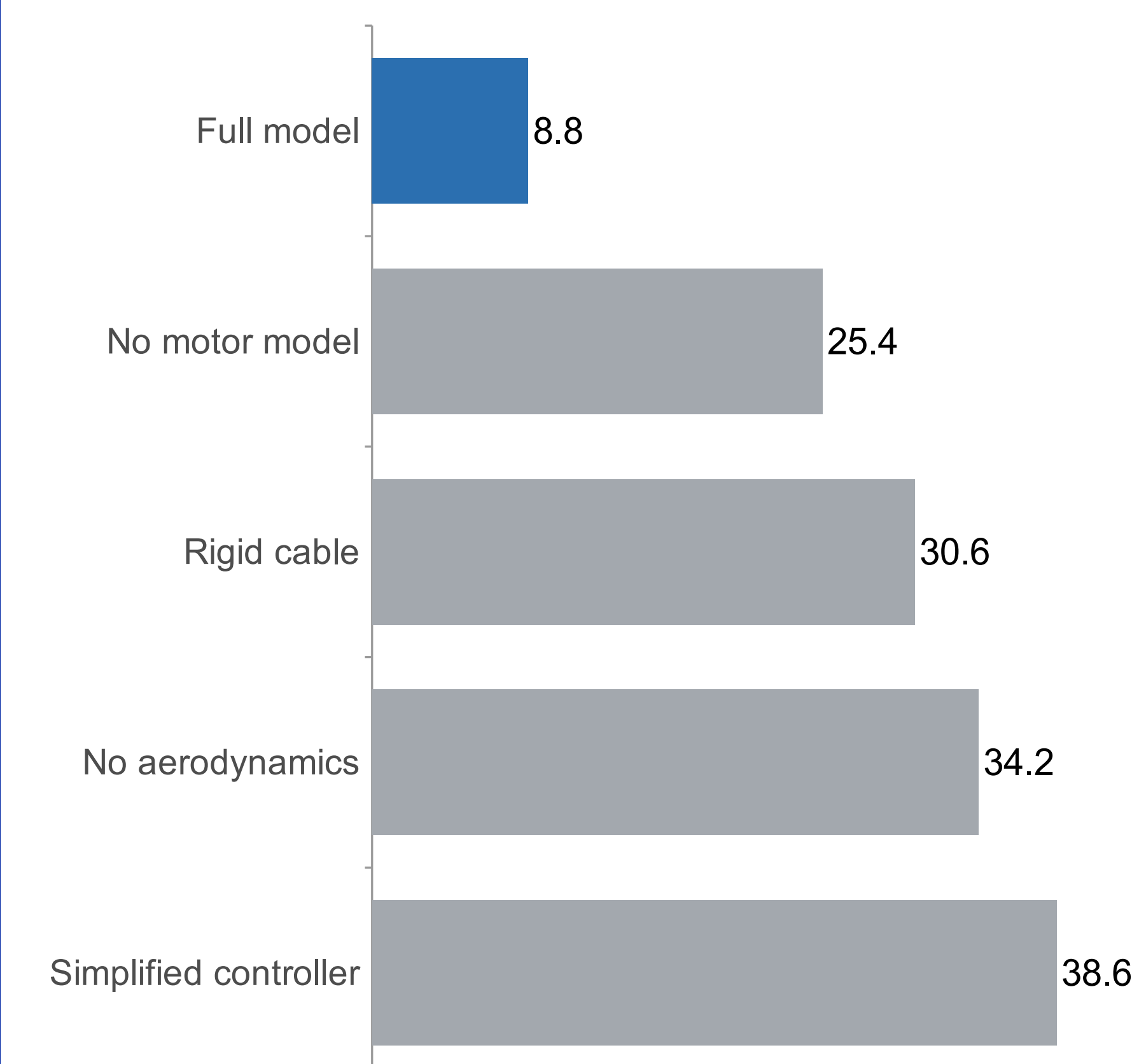
> 80 %

of racing peak thrust and body rate, in a 0.4 s swing

< 60 %

of our peak thrust used by the model-based baselines

Hardware ablations at the 2.0 m target



Removing any one component raises the landing error by a factor of **2.9 to 4.4** ($p \leq 0.0007$).